MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1 2025/2026

PROGRESS REPORT #3

MICROCONTROLLER-BASED AUTOMATIC WASHING MACHINE WITH LOAD DETECTION

SECTION 2

GROUP 16

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INTRODUCTION

This progress report documents the activities carried out during Week 12, focusing on system assembly and data testing of the automated washing machine project. At this stage, all major hardware components, including sensors, indicators, display units, and control buttons, have been integrated into a single working system. The objective of this phase is to verify that the system operates according to the designed workflow, responds correctly to user inputs, and executes each washing cycle in the correct sequence and timing.

System assembly and testing are critical to ensure that individual modules, such as the weight sensor, LCD display, LEDs, timer, and motor control, work reliably when combined. This report outlines the progress achieved, the testing procedures conducted, and observations gathered during real-time operation of the system.

SYSTEM ASSEMBLY PROCESS

During this week, the washing machine system was fully assembled according to the designed system architecture and flowchart. The following integrations were completed successfully:

- The **weight sensor** was installed and calibrated to detect the total load of clothes placed inside the machine.

- The **LCD display** was connected to the controller to present real-time information, including detected weight, suggested water quantity, filling time, and system status messages such as *"DRAIN"* and *"DONE"*.

- **Push buttons** (Start, Stop, and Reset) were connected and tested for correct logic operation.

- **Green and Red LEDs** were installed to provide visual feedback on motor operation status.

- The **motor control system** was wired and synchronized with timer-based logic to control washing, spinning, and rinsing processes.

All components were mounted securely, and wiring connections were verified to prevent loose contacts or incorrect pin assignments.

DATA TESTING PROCEDURE

Data testing was conducted by operating the system through multiple complete washing cycles. The following step-by-step sequence was tested and verified:

1. Clothes were loaded into the machine.
2. The **weight sensor** detected the total load accurately.
3. The **LCD display** showed:
    - Total weight of clothes
    - Suggested water quantity
    - Estimated filling time
4. Upon pressing the **Start button**, the washing process began.
5. The **Green LED** turned ON, indicating motor rotation.
6. A **40-second timer** was activated to simulate the washing duration.
7. After washing, the system automatically transitioned to the **spinning process**.
8. The **Green LED** remained ON, confirming continued motor operation.
9. Another **40-second timer** controlled the spinning phase.
10. The system then proceeded to the **rinsing process**.
11. The **Green LED** remained ON during rinsing.
12. A third **40-second timer** completed the rinsing phase.
13. After rinsing, the LCD displayed **“DRAIN”**, indicating water drainage.
14. After **20 seconds**, the LCD displayed **“DONE”**, confirming completion.
15. Pressing the **Stop button** halted the system immediately at any stage.
16. Pressing the **Reset button** returned the system to the display stage (Step 3).
17. The **Red LED** turned ON whenever the motor was not rotating.

Each process transition was observed to ensure correct timing, display updates, and LED indications.

OBSERVATION AND DATA RESULT

- The weight sensor consistently detected changes in load with stable readings.
- LCD messages updated correctly at each stage without delay.
- Timer-controlled processes executed for the intended durations.
- The system correctly transitioned between washing, spinning, rinsing, draining, and completion stages.
- Emergency controls (Stop and Reset buttons) functioned as intended.
- LED indicators accurately reflected motor status throughout operation.

No critical system failures were observed during testing, indicating that the integrated system operates reliably.

ISSUES ENCOUNTERED AND IMPROVEMENT

Minor delays were observed during LCD refresh at process transitions, which can be improved through software optimization. Additionally, future testing may include variable timing based on load weight to further enhance system efficiency.

CONCLUSION

The **system assembly and data testing phase** was successfully completed in Week 12. All hardware components were integrated properly, and the system demonstrated stable and predictable behavior during multiple test cycles. The washing machine control system operates according to the designed logic and flow, meeting the objectives of this project phase. The system is now ready for final validation, documentation, and presentation.

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been untaken or done by unspecified sources or persons. We hereby certify that this report has **not been done by only one individual,** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have **read** and **understand** the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for **marking,** and this **final printed report** has been **verified by us.**

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